

N. Deklen Crocker

Electrical Engineer (E.I.T)

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🏠 Edmonton, AB

in [/ndcrocker](#)

🌐 [/NDCrocker](#)

Experience

Jones Microwave Inc.

Aug. 2022 – Present

Embedded System Developer

Edmonton, AB

- › Develop a comprehensive market survey of several laser diode drivers for viability in custom PCB design
- › Compare physical attributes of VCSELs and laser diodes for most appropriate device selection in project
- › Contact component suppliers and producers for further inquiries into their products
- › Create flexible, low-cost and highly tunable circuit designs for bias and control of mmWave guide switches
- › Devise comprehensive testing plans to evaluate completed circuit and PCB Designs
- ›
- › **Technologies:** KiCAD, LTSpice, Laser Diodes, LED Drivers, Keysight Oscilloscopes,

Government of the Northwest Territories

Apr. 2019 – Aug. 2021

Project Assistant (Summer Internship)

Inuvik, NT

- › Build and upgrade VBA tools in Excel and Access for Project Manager to receive the latest project information from all project officers across our office's jurisdiction
- › Maintain project budget and status data for over 100 projects in the Beaufort and Sahtu regions
- › Complete audits of multi-year and multi-contractor projects to assist in settling contract disputes
- › Compile project drawing database for 1100+ engineering drawings and schematics from the region
- › **Technologies:** Microsoft Access, Microsoft Excel, SQL, Visual Basic for Applications (VBA)

Education

Bachelor of Science in Electrical Engineering

Sep. 2018 – Nov. 2022

University of Alberta

Edmonton, AB

- › Relevant Coursework: Digital Electronics, Embedded System Design, Integrated Circuit Design

Skills

Programming Languages C, C++, Dart, LaTeX, MATLAB, Python, SQL, VBA, VHDL

Engineering Design Tools Cadence Virtuoso, KiCAD, Logic Analyzers, LTSpice, Qucs Studio

Organizational Software Git, Google Drive Suite, MS Office Suite (Word, Excel, Powerpoint, Access)

Projects

Classroom IoT Carbon Dioxide Sensor (Capstone)

Sep. 2021 – Apr. 2022

University of Alberta

Edmonton, AB

- › Design a battery recharging circuit using Lithium-Polymer and NiMH battery chemistries for portable device
- › Develop a smartphone application to interface with sensor to provide a more fluid and controlled UI
- › Frame a Bluetooth GATT Profile to exchange credential data between smartphone application and device
- › Coordinate regular meetings between the group, the client, and technical advisors through all project stages
- › **Technologies:** Bluetooth Low Energy (BLE), C++, Dart, Flutter, KiCAD, SoC

Extracurriculars

University of Alberta Powerlifting Club

Feb. 2022 – Present

- › Plan and schedule social events with other members of the club
- › Increase the community size by marketing club perks and activities to entice individuals to join

AlbertaLoop

Jun. 2022 – Present

- › Create Python script to parse data from navigation module test to diagnose issues within the GPS module's equirectangular projection process
- › **Technologies:** Python

Interests

Personal

- › Learning Languages: DELF-certified fluency in French; interested in learning German and Cantonese
- › Powerlifting: Avid powerlifting competitor
- › Reading: Taking time to reading more often. Current interests: Neurology, Psychology, Electrical Engineering
- › Cooking: Enjoy learning a new recipe and cooking for friends and family
- › Curling: Curled competitively for 10 years; multiple time NWT representative between Nationals and CWGs

References

Available Upon Request